# CO4 - AT2 - INDUSTRY PROBLEM SOLVING

**1.A telecommunications company wants to recommend suitable data plans to customers based on their usage patterns. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity between users is measured, how recommendations are generated, and how the system adapts to changing user behavior over time.**

## Data Plan Recommendation Using K-Nearest Neighbour

## Introduction

A telecommunications company can use **K-Nearest Neighbour (KNN)** to recommend suitable data plans based on customers' usage patterns. KNN is a supervised learning algorithm that recommends a plan by comparing a new customer's behaviour with similar existing customers.

## Input Features

The system can consider the following customer usage parameters:

- Monthly data consumption (GB)
- Number of voice calls
- SMS usage
- Average call duration
- Number of active days
- Current monthly bill
- Usage during peak and off-peak hours

## Measuring Similarity

Similarity between customers can be measured using **Euclidean distance**. For two customers, the distance can be calculated as:

**Distance** $= \sqrt{[(x_1-y_1)^2 + (x_2-y_2)^2 + \dots + (x_n-y_n)^2]}$

A smaller distance means that the two customers have more similar usage patterns.

Before calculating the distance, the features should be normalized because data usage, call counts, and billing amounts may have different scales.

**Recommendation Process**

1. Collect historical customer usage and their selected data plans.
2. Normalize the usage features.
3. Select a value of **K**, such as 5.
4. Calculate the distance between the new customer and existing customers.
5. Identify the K most similar customers.
6. Find the most commonly used plan among these neighbours.
7. Recommend that plan to the new customer.
8. The system can also display alternative plans based on customer requirements.

**Adaptation to Changing Behaviour**

KNN can adapt easily because it does not require a complex retraining process. New customer usage records can continuously be added to the database. If a customer's behaviour changes—for example, their monthly data consumption increases—their latest usage can be used for future recommendations.

A **recent-usage weighting** method can also be used so that recent behaviour has more influence than older behaviour.

**Example**

Suppose a customer uses approximately **25 GB of data per month**, makes 300 calls, and spends ₹500 per month. The system finds five existing customers with similar usage. If most of those customers use a **30 GB data plan**, the system recommends the 30 GB plan.

**Advantages**

- Simple and easy to implement.
- Works well with customer usage data.
- Adapts to new customers and changing behaviour.
- No complex model training is required.
- Can provide personalized recommendations.

**Conclusion**

KNN provides an effective solution for telecom data-plan recommendation by identifying customers with similar usage patterns and recommending the plan most suitable for them. Continuous updating of customer data allows the system to adapt to changing usage behaviour.

**2. An energy management company aims to predict electricity consumption based on factors such as time, weather, and appliance usage. Propose a solution using Locally Weighted Regression or RBF networks. Explain how local learning captures variations in consumption patterns and how the model handles noisy and seasonal data.**

**Electricity Consumption Prediction Using Locally Weighted Regression**

**Introduction**

An energy management company can use **Locally Weighted Regression (LWR)** to predict electricity consumption based on factors such as time, weather, and appliance usage. Unlike ordinary regression, LWR gives greater importance to data points that are close to the current input.

**Input Features**

The prediction model can use:

- Time of day
- Day of the week
- Temperature
- Humidity
- Weather conditions
- Number of appliances in use
- Previous electricity consumption
- Holiday or working-day information

**Local Learning**

LWR performs prediction by considering nearby historical observations. For a particular time and weather condition, data points with similar conditions receive higher weights.

For example, when predicting electricity consumption at **7 PM on a hot day**, the model gives more importance to previous observations recorded around 7 PM on similarly hot days.

The weight can be represented using a Gaussian function:

$$\mathbf{w_i = \exp[-(x - x_i)^2 / (2\tau^2)]}$$

where:

- $\mathbf{w_i}$ = weight of the data point
- $\mathbf{x}$ = current input
- $\mathbf{x_i}$ = historical data point
- $\boldsymbol{\tau}$ = bandwidth controlling the locality

**Handling Consumption Variations**

Electricity usage changes throughout the day. For example:

- Low consumption may occur during early morning.

- Consumption increases during working hours.
- High consumption may occur during evening hours.
- Air-conditioner usage may increase consumption during hot weather.

LWR captures these variations because it learns from nearby and relevant observations instead of applying one global relationship to all data.

### Handling Noisy Data

Real-world electricity data may contain noise caused by sensor errors, sudden appliance usage, or unexpected events. LWR reduces the influence of unrelated observations by assigning them smaller weights.

Data preprocessing such as removing extreme outliers and normalizing features can further improve prediction accuracy.

### Handling Seasonal Data

Electricity consumption can change with seasons:

- **Summer:** Higher air-conditioner and cooling usage.
- **Winter:** Higher heating or other seasonal appliance usage.
- **Monsoon:** Weather-dependent changes in appliance usage.
- **Weekends:** Different consumption patterns compared with weekdays.

LWR can capture these patterns by using historical observations that are similar in time, weather, and usage conditions.

### Example

Suppose the company wants to predict electricity consumption for **8 PM on a hot weekday**. LWR searches historical data for observations close to 8 PM, with similar temperatures and weekday conditions. These observations receive higher weights, and the model uses them to estimate the expected electricity consumption.

**Advantages**

- Captures local variations in electricity usage.
- Handles nonlinear relationships effectively.
- Adapts to different time and weather conditions.
- Gives less importance to distant or unrelated observations.
- Useful for changing consumption patterns.

**Conclusion**

Locally Weighted Regression is suitable for electricity-consumption prediction because it focuses on similar historical conditions. By giving higher weights to relevant observations, it can capture daily, weather-related, and seasonal variations while reducing the effect of noisy data.